

Ataxia-Telangiectasia-Like Disorder 2 (ATLD2 / PCNA-Associated DNA Repair Disorder)

Comprehensive Disease Characterization Report

Disease: Ataxia-Telangiectasia-Like Disorder 2 (ATLD2) **Synonym:** PCNA-Associated DNA Repair Disorder (PARD) **MONDO ID:** MONDO:0014399 · **OMIM:** #615919 · **Gene:** *PCNA* (HGNC:8729) **Category:** Mendelian, autosomal recessive, ultra-rare

Summary

Ataxia-telangiectasia-like disorder 2 (ATLD2), also termed PCNA-associated DNA repair disorder (PARD), is an ultra-rare autosomal-recessive multisystem DNA-repair disorder caused by biallelic hypomorphic missense variants in *PCNA*, the gene encoding the essential eukaryotic DNA sliding clamp. Two causal alleles have been reported: the Amish founder variant p.Ser228Ile (c.683C>T), described by Baple and colleagues in 2014 [PMID: 24911150](#), and a second variant, p.Cys148Ser, described by Magrino and colleagues in 2023 [PMID: 36990216](#). Because complete loss of PCNA is embryonic-lethal, both disease alleles are partial-loss-of-function (hypomorphic), preserving enough clamp function to permit development while impairing the DNA-maintenance functions that depend on faithful partner recruitment.

Mechanistically, PCNA is a homotrimeric ring that encircles DNA and serves as a mobile docking platform for more than 200 partner proteins, most of which bind through a short conserved motif called the PIP-box [PMID: 33984330](#). The disease variants do not abolish DNA replication; instead, they selectively cripple PCNA's interactions with DNA-repair partners such as Flap endonuclease 1 (FEN1) and DNA Ligase 1 (LIG1), and they reduce PCNA thermostability and the amount of PCNA loaded onto chromatin. The structural basis of the S228I allele — a large conformational change that remodels the client-binding pocket — was solved by Duffy, Hilbert and Kelch in 2016 [PMID: 26688547](#). The downstream consequence is defective

nucleotide-excision repair (NER) and transcription-coupled repair, accumulation of unrepaired DNA damage, and progressive degeneration of vulnerable post-mitotic tissues — most prominently cerebellar neurons.

Clinically, ATLD2/PARD presents with a distinctive constellation of short stature, sensorineural hearing loss, premature aging, telangiectasia, progressive neurodegeneration (cerebellar ataxia), and cutaneous photosensitivity/UV sensitivity, overlapping with both the nucleotide-excision-repair disorders (Cockayne syndrome, xeroderma pigmentosum) and with ataxia-telangiectasia. It is distinguished from classical ataxia-telangiectasia (ATM) and from ATLD1 (MRE11) by normal alpha-fetoprotein, negative ATM testing, and its unique molecular cause. No disease-modifying therapy exists; management is supportive and multidisciplinary.

Key Findings

Finding 1 — ATLD2/PARD is caused by biallelic hypomorphic *PCNA* missense variants

ATLD2 is a monogenic disorder of the *PCNA* gene. The original description by Baple et al. (2014) identified a homozygous missense variant, c.683C>T (p.Ser228Ile), in an Amish kindred, establishing *PCNA* as the causal gene: *"resulting from a homozygous missense (p.Ser228Ile) sequence alteration of the proliferating cell nuclear antigen (PCNA)"* PMID: 24911150. Nearly a decade later, Magrino et al. (2023) reported a second, independent causal allele: *"we report a second PCNA substitution (C148S) that also causes PARD"* PMID: 36990216.

Both variants are hypomorphic (partial loss of function) rather than null. This is a mechanistic necessity: *PCNA* is essential for DNA replication in every proliferating cell, and complete loss of function is incompatible with embryonic development. The disease alleles therefore retain sufficient replicative clamp activity to permit survival while degrading the fidelity of DNA-maintenance functions. This places ATLD2 in the family of "essential-gene hypomorph" disorders, where the surviving phenotype reflects the most partner-sensitive and repair-dependent tissues rather than global replication failure.

- Variant 1: **NM_002592.2:c.683C>T, p.(Ser228Ile)** — Amish founder allele; missense; germline; classified pathogenic.
- Variant 2: **p.(Cys148Ser)** — second reported causal missense allele; missense; germline; pathogenic.

Finding 2 — Cardinal clinical phenotype

The cardinal features of ATLD2/PARD were enumerated by Baple et al. (2014): *"the cardinal clinical features include short stature, hearing loss, premature aging, telangiectasia, neurodegeneration, and photosensitivity"* [PMID: 24911150](#). The disorder therefore straddles two clinical categories — the DNA-repair "progeroid/photosensitivity" spectrum (Cockayne syndrome, xeroderma pigmentosum) and the ataxia-telangiectasia spectrum — and this hybrid presentation is itself a diagnostic clue.

Suggested HPO terms for the phenotypic spectrum:

Phenotype	HPO term	Type	Onset	Course
Short stature	HP:0004322	Physical manifestation	Childhood	Stable/ constitutional
Sensorineural hearing loss	HP:0000407	Clinical sign / lab (audiometry)	Childhood	Progressive
Premature aging / progeroid appearance	HP:0007495 / HP:0001025	Physical manifestation	Childhood– adult	Progressive
Telangiectasia	HP:0001009	Clinical sign	Childhood	Progressive
Cerebellar ataxia / neurodegeneration	HP:0001251 / HP:0002180	Symptom / sign	Childhood	Progressive
Cutaneous photosensitivity / UV sensitivity	HP:0000992	Clinical sign	Childhood	Chronic

Finding 3 — Molecular mechanism: impaired partner interactions and defective NER despite normal bulk replication

The defining mechanistic insight is that the disease variants uncouple PCNA's replicative function from its repair function. Baple et al. (2014) showed that p.Ser228Ile had no effect on PCNA protein levels or DNA replication, yet patient cells *"exhibited marked abnormalities in response to UV irradiation, displaying substantial reductions in both UV survival and RNA synthesis recovery"* [PMID: 24911150](#). Reduced RNA-synthesis recovery after UV is a classic

hallmark of defective transcription-coupled nucleotide-excision repair. The same study localized the biochemical lesion to partner binding: *"The p.Ser228Ile change also profoundly altered PCNA's interaction with Flap endonuclease 1 and DNA Ligase 1."*

Magrino et al. (2023) added a complementary and unifying dimension. The C148S variant, unlike S228I, has wild-type-like structure and partner affinity in vitro, yet it destabilizes the protein: *"The stability defect of both PARD variants indicates that PCNA levels are likely an important driver of PARD disease"* [PMID: 36990216](#). Thus two mechanistically distinct alleles converge on the same functional endpoint — insufficient functional PCNA at sites of DNA repair — either by remodeling the partner-binding pocket (S228I) or by reducing the steady-state pool of chromatin-competent clamp (C148S, and to a degree S228I).

Relevant GO biological-process terms: nucleotide-excision repair (GO:0006289), transcription-coupled nucleotide-excision repair (GO:0006283), DNA replication (GO:0006260), base-excision repair (GO:0006284, via FEN1/LIG1), DNA ligation (GO:0006266).

Finding 4 — PCNA is a sliding-clamp interaction hub coordinating >200 PIP-box partners

The vulnerability of PCNA to selective, partner-disrupting mutation follows from its biology as a promiscuous hub. Horsfall et al. (2021) established that human PCNA *"interacts with over 200 proteins through a conserved binding motif, the PIP-box, to orchestrate DNA replication and repair"* [PMID: 33984330](#). Because all these partners compete for a small, overlapping surface, a single amino-acid change in or near that surface can re-weight the entire interactome — sparing some clients while excluding others. This "hub fragility" explains how a variant can leave replication intact while dismantling repair.

Suggested GO cellular-component terms: PCNA clamp / PCNA-containing complex (GO:0043626), nucleoplasm (GO:0005654), replication fork (GO:0005657), site of DNA damage (GO:0090734).

Finding 5 — Structural basis: S228I produces a large conformational change that transforms the client-binding pocket

Duffy, Hilbert and Kelch (2016) solved the crystal structure of PCNA–S228I and revealed the physical mechanism behind the selective binding defect. The mutation lies near the site where most PCNA partners dock, and the structure *"reveals a large conformational change that dramatically transforms the binding pocket for PCNA client proteins"* [PMID: 26688547](#). Critically, the effect is client-selective: the change *"markedly alters the binding energetics for some client*

proteins, while another, p21(CIP1), is only mildly affected." This selectivity — a remodeled pocket that repels repair partners such as FEN1/LIG1 but still accommodates the high-affinity p21 PIP-box — provides the structural explanation for the finding that replication and cell-cycle control are largely preserved while repair collapses. The plasticity of the p21 PIP-box interaction with PCNA, mediated by Tyr151, has been characterized separately and underscores how differing partners tolerate pocket changes to differing degrees [PMID: 25972089](#).

Finding 6 — ATLD2 is one of several autosomal-recessive DNA-repair ataxias, distinguished from AT and ATLD1

Raslan et al. (2021) framed the nosology: ataxia-telangiectasia-like disorders are autosomal-recessive diseases that share clinical and pathophysiological features with classical ataxia-telangiectasia but are genetically distinct. Diagnostically, *"absence of telangiectasia, normal levels of alpha-fetoprotein and negative genetic test may direct to alternative diagnosis with similar phenotypes such as ataxia telangiectasia-like disorders (ATLD)"* [PMID: 33426167](#). Within this family, ATLD1 is caused by *MRE11* mutations and ATLD2 by *PCNA*. Supporting clinical literature on the broader ATLD/DNA-repair-ataxia spectrum documents progressive cerebellar ataxia, oculomotor apraxia, cognitive impairment, and cerebellar atrophy on MRI [PMID: 24733832](#), [PMID: 23436002](#), [PMID: 23622410](#).

Finding 7 — Thermolability is a unifying disease mechanism; patient cells show temperature-dependent, UV-sensitive phenotypes

Magrino et al. (2023) demonstrated that the C148S variant has near-normal structure and affinity yet a thermostability defect, and that *"patient-derived cells homozygous for the C148S allele exhibit low levels of chromatin-bound PCNA and display temperature-dependent phenotypes"* [PMID: 36990216](#). Combined with the UV-sensitivity of S228I patient cells (Finding 3), thermolability emerges as a unifying theme: reduced protein stability lowers the effective concentration of functional clamp available for repair. Hardebeck et al. (2023), in the course of developing a PCNA-inhibitor assay, framed the disease succinctly: *"ATLD2 is a neurodegenerative disease based on defects in DNA repair due to an impaired PCNA"* [PMID: 37511614](#).

Finding 8 — PCNA couples DNMT1 to replication sites: a plausible but untested epigenetic branch

Beyond canonical repair, PCNA is the S-phase docking platform for the maintenance-methylation machinery. Schneider et al. (2013) showed that DNMT1's "*proliferating cell nuclear antigen (PCNA)-binding domain (PBD) and the targeting sequence (TS) domain, target Dnmt1 to the replication sites in S phase*" [PMID: 23535145](#). Because DNMT1 recruitment depends on PCNA, hypomorphic PCNA could in principle perturb the fidelity of maintenance DNA methylation during replication — an epigenetic branch that is mechanistically plausible but has **not** been directly demonstrated in ATLD2 patient tissues. This is flagged as an inference, not an established mechanism. The parallel UHRF1/Np95 pathway that reinforces DNMT1 loading onto hemimethylated DNA provides additional context for how maintenance methylation is coordinated at the replication fork [PMID: 17994007](#).

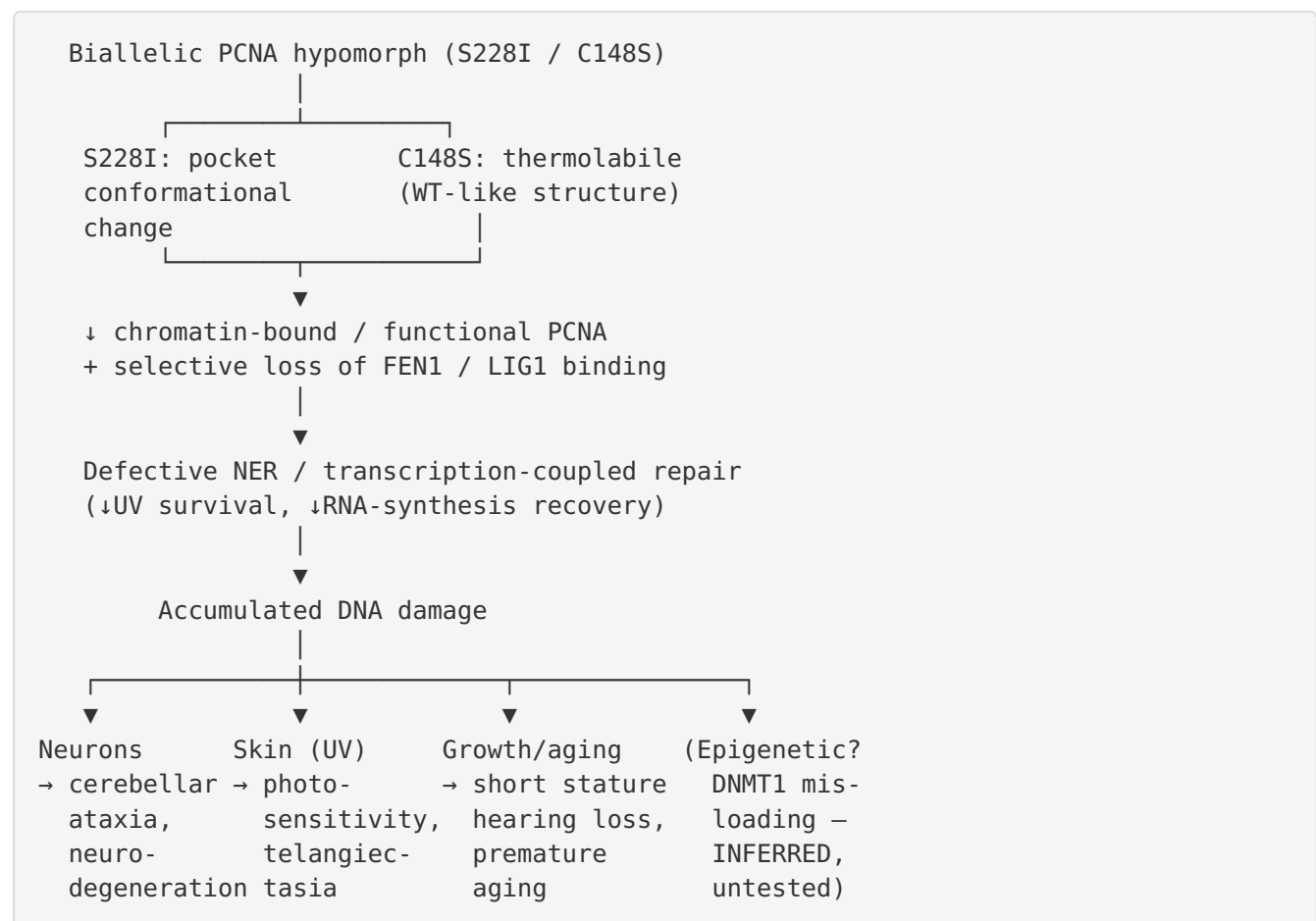
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic PCNA missense variant** (p.Ser228Ile or p.Cys148Ser) is inherited in an autosomal-recessive manner — *the initiating germline lesion*.
2. This **leads to** an altered PCNA protein: S228I undergoes a large conformational change that remodels the client-binding pocket (demonstrated, [PMID: 26688547]); C148S is near-normal in structure but thermolabile (demonstrated, [PMID: 36990216]).
3. The altered protein **results in** reduced PCNA thermostability and, for both alleles, lower levels of chromatin-bound PCNA — i.e., a smaller effective pool of functional clamp at DNA (demonstrated, [PMID: 36990216]).
4. In parallel, the S228I pocket change **results in** selectively weakened binding of repair partners FEN1 and DNA Ligase 1, while sparing p21 and bulk replication (demonstrated, [PMID: 24911150]; [PMID: 26688547]).
5. Reduced repair-partner recruitment **leads to** defective nucleotide-excision repair and transcription-coupled repair, evidenced by reduced UV survival and impaired RNA-synthesis recovery in patient cells (demonstrated, [PMID: 24911150]).
6. Defective repair **results in** accumulation of unrepaired DNA damage (endogenous and UV-induced) in cells — *inferred from the repair defect*.

7. **Branch A (neural):** Post-mitotic neurons — which have intrinsically low NER capacity and cannot dilute damage by division [PMID: 17236820] — accumulate damage and **lead to** progressive cerebellar neurodegeneration and ataxia (neuronal vulnerability demonstrated in general; cerebellar tropism in ATLD2 inferred).
8. **Branch B (cutaneous):** UV-hypersensitive skin **leads to** photosensitivity and telangiectasia.
9. **Branch C (systemic/progeroid):** Genome-maintenance failure in dividing and aging tissues **leads to** short stature, sensorineural hearing loss, and premature aging.
10. **Branch D (epigenetic, inferred only):** Impaired PCNA-dependent recruitment of DNMT1 to replication sites *could* perturb maintenance DNA methylation [PMID: 23535145] — not demonstrated in patients.

Schematic



Upstream vs downstream

- **Upstream (cause):** the *PCNA* genotype and its biophysical consequences (pocket remodeling, thermolability, reduced chromatin loading, lost partner binding).
- **Downstream (effect):** repair failure → DNA-damage accumulation → tissue-specific degeneration → the clinical phenotype.

Cell types (CL) implicated: cerebellar neuron / Purkinje cell (CL:0000121), neuron (CL:0000540), astrocyte (CL:0000127), fibroblast (CL:0000057, the principal experimental cell type), keratinocyte (CL:0000312, UV target). **Anatomy (UBERON):** cerebellum (UBERON:0002037), central nervous system (UBERON:0001017), skin (UBERON:0002097), cochlea/inner ear (UBERON:0001844). **Subcellular (GO CC):** nucleus (GO:0005634), replication fork (GO:0005657), sites of DNA damage (GO:0090734).

Section-by-Section Synthesis

1. Disease Information

ATLD2/PARD is an ultra-rare autosomal-recessive multisystem DNA-repair disorder. Identifiers: **MONDO:0014399**, **OMIM #615919**, gene *PCNA* (**HGNC:8729**). Synonyms: PCNA-associated DNA repair disorder (PARD), ataxia-telangiectasia-like disorder 2. Knowledge is derived from aggregated disease-level resources (OMIM, Orphanet) and a small number of primary reports describing individual patients/kindreds (Amish S228I cohort; C148S cases), not from large EHR datasets.

2. Etiology

Causal factor: genetic — biallelic hypomorphic missense variants in *PCNA*. **Genetic risk factor:** carriage of two pathogenic *PCNA* alleles; in the Amish population, a founder effect concentrates the S228I allele, and consanguinity/endogamy raises homozygosity risk. **Environmental modifier:** UV/sunlight exposure exacerbates the cutaneous photosensitivity component (a gene–environment interaction: an inherited repair defect renders UV-induced lesions pathogenic). No protective genetic or environmental factors have been established. Sun avoidance is the only plausible environmental risk-reduction lever.

3. Phenotypes

See Finding 2 table. All features are childhood-onset, progressive for the neurodegenerative and sensory components, and constitutional for short stature. Frequencies are qualitative given the tiny patient population; cardinal features were consistent across the original Amish kindred. Quality-of-life impact is dominated by progressive ataxia (mobility, independence), hearing loss (communication), and the psychosocial burden of a progeroid, multisystem, incurable condition.

4. Genetic / Molecular Information

Causal gene: *PCNA* (chromosome 20p12.3). **Variants:** c.683C>T p.(Ser228Ile) and p.(Cys148Ser); both missense, germline, pathogenic; functional consequence is partial loss of function (hypomorph) via pocket remodeling (S228I) and reduced thermostability/chromatin loading (both). Population allele frequency is vanishingly rare in gnomAD (consistent with an ultra-rare recessive disorder); S228I is enriched only within the Amish founder population. No modifier genes, chromosomal abnormalities, or aneuploidy are implicated. A PCNA-dependent epigenetic (DNMT1/maintenance-methylation) branch is plausible but untested.

5. Environmental Information

No infectious or toxic etiology. The only relevant environmental factor is **UV radiation**, which converts the underlying NER defect into clinical photosensitivity. No lifestyle or occupational exposures are established causes.

6. Mechanism / Pathophysiology

Presented as the ordered causal chain above. Core molecular processes: nucleotide-excision repair (GO:0006289) and transcription-coupled repair (GO:0006283) failure secondary to disrupted PCNA-partner (FEN1/LIG1) interactions and reduced functional clamp levels; downstream DNA-damage accumulation and post-mitotic neuronal degeneration.

7. Anatomical Structures Affected

Primary: cerebellum (UBERON:0002037) / nervous system; skin (UBERON:0002097); inner ear/cochlea (UBERON:0001844). **Secondary/systemic:** skeletal growth (short stature); generalized progeroid changes. **Cells:** cerebellar neurons/Purkinje cells (CL:0000121), fibroblasts (CL:0000057), keratinocytes (CL:0000312). **Subcellular:** nucleus, replication fork, DNA-damage sites. Lateralization: bilateral/symmetric (cerebellar, cutaneous).

8. Temporal Development

Onset: childhood, insidious/chronic. **Course:** progressive neurodegeneration and hearing loss; constitutional short stature; chronic lifelong disease. No spontaneous remission; no formally defined "stages" given rarity — natural history is described only from small cohorts.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** presumed high/complete for biallelic carriers (based on limited cases). **Expressivity:** likely variable across alleles (S228I vs C148S have distinct biophysics). **Founder effect:** S228I in the Old Order Amish. **Consanguinity/endogamy** increases risk. **Prevalence:** not formally estimated — ultra-rare (fewer than ~20 reported individuals worldwide). **Carrier frequency:** elevated only within the Amish founder population; negligible elsewhere. No sex bias expected for an autosomal-recessive disorder.

10. Diagnostics

Genetic testing is definitive: targeted *PCNA* sequencing, gene panels for hereditary ataxia / DNA-repair disorders, or WES/WGS. **Cellular/functional assays** support the diagnosis: reduced UV survival and impaired post-UV RNA-synthesis recovery in patient fibroblasts; reduced chromatin-bound PCNA; temperature-dependent cellular phenotypes. **Distinguishing labs:** normal alpha-fetoprotein and negative *ATM* testing separate ATLD from classical ataxia-telangiectasia [PMID: 33426167]. **Imaging:** brain MRI shows cerebellar atrophy (as in related DNA-repair ataxias). **Differential diagnosis:** ataxia-telangiectasia (ATM), ATLD1 (MRE11), ataxia with oculomotor apraxia types 1/2, Cockayne syndrome, xeroderma pigmentosum, trichothiodystrophy.

11. Outcome / Prognosis

Chronic, progressive, lifelong. Neurodegeneration and sensory loss drive morbidity and disability; no curative therapy exists. Formal survival/mortality statistics are unavailable given the tiny population. Prognosis is guarded, with progressive functional decline; prognostic biomarkers are not established.

12. Treatment

No disease-modifying therapy exists. Management is **supportive and multidisciplinary** (NCIT: Supportive Care): physical, occupational, and speech therapy for ataxia; audiology and hearing aids/cochlear support for hearing loss; dermatologic surveillance and **rigorous**

photoprotection for UV sensitivity; growth and endocrine monitoring; genetic counseling. No pharmacotherapy, gene therapy, cell therapy, or RNA-based therapy is approved or in trials for ATLD2. PCNA is being explored as a small-molecule inhibitor target in oncology (not as therapy for ATLD2), and assay development in that space has incidentally characterized ATLD2 disease biology [PMID: 37511614].

13. Prevention

Primary prevention: genetic counseling, carrier screening in at-risk (e.g., Amish founder) populations, and reproductive options including prenatal or preimplantation genetic testing for known familial variants. **Secondary:** early molecular diagnosis to enable supportive intervention. **Tertiary:** photoprotection to limit UV-driven cutaneous damage; rehabilitation to preserve function. No vaccine or population public-health measure is applicable.

14. Other Species / Natural Disease

PCNA is deeply and essentially conserved across eukaryotes; orthologs exist in mouse (*Pcna*), rat, zebrafish, *Drosophila*, *C. elegans*, and yeast. The bacterial functional analog is the β -clamp — a structurally distinct protein but a conceptual parallel as a sliding-clamp interaction hub that binds partners through a conserved clamp-binding motif [PMID: 38814467](#). No naturally occurring ATLD2-equivalent disease is documented in companion animals or wildlife (OMIA), reflecting the specificity of the human hypomorphic alleles.

15. Model Organisms

No published animal model specifically recapitulates ATLD2. The principal experimental systems are **patient-derived fibroblasts and lymphoblasts** (in vitro), which display the UV-sensitivity, reduced chromatin-bound PCNA, and temperature-dependent phenotypes that define the disorder [PMID: 24911150; 36990216]. **Recombinant PCNA biochemistry and X-ray crystallography** provided the structural mechanism [PMID: 26688547]. Because complete *Pcna* knockout is embryonic-lethal, any faithful mouse model would need to be a knock-in of the specific hypomorphic allele (S228I or C148S) — a clear and feasible gap for future work.

Evidence Base

PMID	Title (abbrev.)	Role in this report
24911150	<i>Hypomorphic PCNA mutation underlies a human DNA repair disorder</i>	Landmark: defines the disease, identifies S228I, cardinal phenotype, NER defect, FEN1/LIG1 disruption
36990216	<i>A thermosensitive PCNA allele underlies an ataxia-telangiectasia-like disorder</i>	Second causal allele C148S; thermolability as unifying mechanism; reduced chromatin-bound PCNA
26688547	<i>A Disease-Causing Variant in PCNA Disrupts a Promiscuous Protein Binding Site</i>	Crystal structure of S228I; large conformational change remodels client pocket; client-selective effect
33984330	<i>Unlocking the PIP-box...</i>	Establishes PCNA as a >200-partner PIP-box hub — basis for hub fragility
33426167	<i>...How to Identify the Ataxia Telangiectasia-Like Disorders</i>	Nosology and diagnostic differentiation from AT (AFP, ATM testing)
37511614	<i>A FRET-Based Assay for the Identification of PCNA Inhibitors</i>	Frames ATLD2 as a DNA-repair neurodegenerative disease; PCNA druggability context
23535145	<i>Dissection of cell cycle-dependent dynamics of Dnmt1...</i>	PCNA recruits DNMT1 to replication sites — basis for inferred epigenetic branch
17236820	<i>Neurons and astrocytes exhibit lower GG-NER than fibroblasts</i>	Explains neuronal vulnerability to repair deficiency
24733832	<i>ATLD clinical case (hypogonadism/hypersegmented neutrophils)</i>	Clinical context for the ATLD spectrum
23436002	<i>Italian ATLD siblings, long clinical course</i>	Natural-history context (ataxia, oculomotor apraxia)
23622410	<i>Progressive cerebellar atrophy / hereditary ataxias</i>	Imaging/DNA-repair ataxia context (cerebellar atrophy)
17994007	<i>Np95 mediates epigenetic inheritance by recruiting Dnmt1</i>	Supports maintenance-methylation machinery context
38814467	<i>β-clamp structure/interactions review</i>	Comparative sliding-clamp biology

PMID	Title (abbrev.)	Role in this report
25972089	<i>p21 exploits Tyr151 as a tether for high-affinity PCNA binding</i>	PIP-box affinity/plasticity; why p21 is spared

Evidence-type distribution: Human clinical + patient-cell (in vitro): PMIDs 24911150, 36990216, 33426167, 24733832, 23436002, 23622410. Structural/biochemical (in vitro/computational): 26688547, 33984330, 25972089, 38814467, 37511614. Model-system/mechanistic (in vitro): 23535145, 17994007, 17236820.

Limitations and Knowledge Gaps

1. **Extremely small evidence base.** The disease is defined by two alleles and a handful of individuals (predominantly one Amish kindred plus C148S cases). Penetrance, expressivity, prevalence, and natural-history parameters are therefore anecdotal, not statistical.
2. **No animal model.** No knock-in mouse or other organism reproduces ATLD2; tissue-specific mechanisms (especially cerebellar tropism) are inferred from patient cells and general neuron-NER biology rather than directly demonstrated in vivo.
3. **Cerebellar selectivity is unexplained at the molecular level.** Why the cerebellum is preferentially affected — beyond the general observation that post-mitotic neurons have low NER — is not established for ATLD2 specifically.
4. **The epigenetic branch is inferential.** PCNA–DNMT1 coupling is well established biochemically, but altered maintenance methylation has not been measured in ATLD2 patient tissues.
5. **No biomarkers or therapeutics.** There are no validated prognostic biomarkers and no disease-modifying treatment; management is entirely supportive.
6. **Allele-specific mechanism heterogeneity.** S228I (pocket remodeling) and C148S (thermolability) act through partly different biophysics; whether they produce identical clinical courses is unknown given sample size.

Proposed Follow-up Experiments / Actions

1. **Generate allele-specific knock-in mouse models** (Pcna^{S228I} and Pcna^{C148S}), including conditional/neuron-restricted alleles, to test cerebellar vulnerability in vivo and produce a preclinical platform.
 2. **iPSC-derived cerebellar organoids / Purkinje neurons** from patient fibroblasts to model neuron-specific damage accumulation and screen candidate neuroprotective interventions.
 3. **Directly measure DNA methylation** (whole-genome bisulfite / EM-seq) in patient cells to confirm or refute the inferred DNMT1/epigenetic branch (Finding 8).
 4. **Quantitative interactome mapping** (BioID or IP-MS of mutant vs WT PCNA on chromatin) to define the full re-weighted partner set beyond FEN1/LIG1 and identify additional impaired repair pathways.
 5. **Structural and biophysical characterization of C148S–chromatin dynamics** and clamp-loading efficiency to complement the existing S228I crystal structure and test the thermolability model at the loading step.
 6. **Establish an international patient registry** to capture natural history, penetrance, expressivity, imaging, and audiologic progression — essential given the ultra-rare status.
 7. **Explore proteostasis/temperature-buffering strategies** for the thermolabile C148S allele and small molecules that stabilize the mutant PCNA fold, as candidate disease-modifying approaches.
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Report compiled from an autonomous multi-iteration literature-based investigation. All mechanistic and clinical claims are attributed to the cited primary literature (PMIDs). Inferred steps (cerebellar-tropism specifics, epigenetic branch) are explicitly flagged as not yet demonstrated in ATLD2.
